

Inference on Smooth Functionals in Dynamic Discrete Choice Models Using Kolmogorov–Arnold Sieve Networks

(Tentative Outline)

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1 Introduction

- Central computational bottleneck in DDCM estimation: dynamic program re-solved at every iteration; high-dimensional latent integration (random coefficients, serially correlated unobservables, common shocks).
- Recap Norets (2009, 2012):
 - MCMC handles integration over latent variables.
 - Feedforward ANNs (FFANNs) approximate $F(s, d, \theta) = \mathbb{E}[V(s_{t+1}; \theta)|s, d; \theta] - \mathbb{E}[V(s_{t+1}; \theta)|s, \bar{d}; \theta]$ offline $\rightarrow$ no per-iteration DP solve.
 - Justification: sup-norm consistency of FFANN sieve (White, 1990; Hornik et al., 1989; Chen, 2007).
- Limitations of FFANN device (Norets, 2012) that KASN resolves:
 - No inference machinery for smooth functionals of F (average derivatives, counterfactual welfare, CCP elasticities).
 - Fixed activations $\rightarrow$ approximation rate opaque to curse of dimensionality as state-parameter dimension grows.
 - Only sup-norm consistency, no rates, no $\sqrt{n}$ -statement for plug-in functionals.
- Contribution – KASN replaces FFANN:
 - Learnable B-spline activations + Group-Lasso ($\ell_{1,2}$) regularization.
 - Dimension-free $o_P(n^{-1/4})$ $L^2(\mu)$ rate over state-parameter space $\rightarrow$ valid posterior inference under MCMC.
 - Three new inference targets:
 - * Average derivative of F w.r.t. continuous states (structural marginal effects).
 - * Counterfactual welfare functionals $\Gamma(\theta_0) = \mathbb{E}[V(s; \theta^{cf})] - \mathbb{E}[V(s; \theta_0)]$ with $\sqrt{n}$ -asymptotic normality.
 - * Average marginal effects on CCPs (Hotz–Miller link), dimension-free behavioral elasticities.
 - Empirical findings (Rust bus engine data, Norets DGPs):
 - * KASN outperforms FFANN in accuracy at comparable sizes.
 - * MCMC posterior closer to exact-DP posterior than FFANN-based.
 - * Nominal coverage for all three functionals; FFANN plug-in misses coverage.

- Structural sparsity: Group-Lasso prunes edges automatically, recovering active state-parameter inputs (latent-factor interpretation).
- Outline: Introduction, Literature, Data, Model, Conclusion.

2 Literature Review

Estimation of Dynamic Discrete Choice Models

- Main literature: **Rust (1987)**; **Hotz and Miller (1993)**; **Aguirregabiria and Mira (2002, 2007)**; Bajari et al. (2007); Su and Judd (2012).
- Identification/unobserved heterogeneity: Magnac and Thesmar (2002); Kasahara and Shimotsu (2009); Arcidiacono and Miller (2011).
- Bayesian/MCMC: Lancaster (1997); Geweke and Keane (2001); Imai et al. (2009); **Norets (2009, 2012)**.
- Surveys: **Aguirregabiria and Mira (2010)**; Keane and Todd (2011); Abbring and Heckman (2007).

DP Solution and Approximation Methods in Estimation

- Curse-of-dimensionality: **Rust (1997)**; **Rust (1996)**.
- Pre-solve/interpolation: Keane and Wolpin (1994); Brown and Flinn (2006); Akerberg (2009); Bajari et al. (2010); Arcidiacono et al. (2013).
- Sparse-grid/polynomial: Krueger and Kubler (2004); Judd (1998).
- Neural-network DP: Maliar et al. (2021); **Fernández-Villaverde et al. (2023)**; Kristensen et al. (2021); Iskhakov et al. (2020).
- Predecessor: Norets (2012) (FFANN for F).

Sieve Estimation and (Artificial) Neural Networks

- Foundations: Grenander (1981); **Hornik et al. (1989)**; **White (1990)**; Barron (1994); Kuan and White (1994).
- Modern sieve theory: Shen (1997); **Chen and Shen (1998)**; **Chen and White (1999)**; **Chen (2007)**; Shen et al. (2023).
- Deep ReLU networks: Schmidt-Hieber (2020, 2021); Yarotsky (2017, 2018); Farrell et al. (2021).
- KART in NN context: Girosi and Poggio (1989); Sprecher (1996, 1997).

Kolmogorov–Arnold Networks and Related Work

- Kolmogorov (1956); Arnold (1957).
- Liu et al. (2024) (KAN); Zhang and Zhou (2024); Gouriéroux and Monfort (2025).
- KASN framework: [self-citation, KASN working paper].

Inference with Machine Learning / Debiased Estimation

- Shen (1997); Chen and Shen (1998); **Ai and Chen (2003)**.
- Average derivative/index: Powell et al. (1989); Ichimura and Lee (1991); Ichimura (1993); Hristache et al. (2001); Horowitz (2009).
- Double/debiased ML: Chernozhukov et al. (2018); Semenova and Chernozhukov (2021); Semenova et al. (2023); Farrell et al. (2025).
- Counterfactual inference: Kasy (2018); Adusumilli et al. (2024); Kalouptsi (2014).

Comparable DDCM Approximation/Inference Approaches

- Norets (2012) (sup-norm only, no functional inference).
- Bajari et al. (2010); Akerberg (2009) (kernel-based, suffer in high dimensions).
- Polynomial/Smolyak (rigid basis).
- Modern deep-learning DP solvers (no inference theory).
- KASN: first dimension-free, $o_P(n^{-1/4})$, functional-inference-ready apparatus for DDCMs.

3 Data

Rust (1987) Bus Engine Replacement Data

- Source: Madison Metropolitan Bus Co. (Harold Zurcher); publicly available via NFXP repository.
- Coverage: 162 buses, 8 subgroups, monthly mileage and replacement decisions (Dec 1974 – May 1985).
- Variables: $x_{t,i}$ (cumulative mileage since last replacement, discretized into $M = 90$ bins); $d_{t,i} \in \{1, 2\}$ (continue / replace); bus group identifier.
- Canonical DDCM benchmark: used directly in Norets (2012); exact DP feasible for EV-IID case $\rightarrow$ ground truth posterior.
- Pre-processing:
 - Reconstruct mileage trajectories, discretize to $M = 90$ bins.
 - CDF-normalize all continuous inputs: $\tilde{X}_{t,j} = \hat{F}_n^j(X_{t,j})$ (required by KASN).
 - Generate training grid: random draws over prior support of θ , following Norets’s procedure.

Simulation Design

- **DGP 1 – EV-IID** (replication of Norets 2012, Sec. 4.4):
 - Utility eq. (4) with EV-IID ε_t , no ν_t .
 - Exact DP via log-sum-exp; artificial panel of $I = 70$ buses, ≈ 4000 points.
 - Compare three estimators: MCMC + exact DP (gold standard), MCMC + FFANN (8-10-1), MCMC + KASN.
- **DGP 2 – Gaussian serially correlated** (Norets 2012, Sec. 4.3):

- $\nu_t \sim N(\rho\nu_{t-1}, h_\nu^{-1})$, trunc.; ε_t truncated normal.
- No exact DP; etalon from random-grid averages ($\tilde{N} = 100$ grids, $\hat{N} = 100$ points).
- Approx. residuals KASN vs. FFANN against etalon.
- Approximation-quality experiments: residual densities, fitted-vs-target plots, sparsity ratio of active edges, wall-clock time per 1000 MCMC draws.
- Estimation experiments: posterior densities of $\alpha_1, \alpha_2, \eta_1, \eta_2$; robustness to KASN hyperparameters (γ, ζ_δ).
- Inference experiments (new):
 - $\Gamma_1(\theta_0) = \mathbb{E}[\partial F / \partial x]$ (coverage, RMSE).
 - $\Gamma_2(\theta_0)$ counterfactual welfare (20% replacement-cost subsidy).
 - $\Gamma_3(\theta_0) = \mathbb{E}[\partial P(d = 2 | X, \theta_0) / \partial x]$ (CCP marginal effect).
 - All compared against FFANN plug-in to document inferential gap.

4 Model

4.1 DDCM Setup (following Norets, 2012)

- State: $s_t = (z_t, x_t, y_t) \in S$ (common unobserved, agent-specific observed, agent-specific unobserved). Decision $d_t \in D$.
- Bellman equations:

$$V(s_t; \theta) = \max_{d_t \in D} \nu(s_t, d_t; \theta), \quad (1)$$

$$\nu(s_t, d_t; \theta) = u(s_t, d_t; \theta) + \beta \mathbb{E}[V(s_{t+1}; \theta) \mid s_t, d_t; \theta]. \quad (2)$$

- Likelihood with high-dimensional latent integration:

$$p(x, d \mid \theta_2) = \int p(x, d, y, \theta_1, z \mid \theta_2) d(y, \theta_1, z), \quad (3)$$

$$\theta = (\theta_1^i, \theta_2).$$

4.2 Bus-Engine Specification

- Per-period utility:

$$u(x_t, \xi_t, \eta_t, d_t; \theta) = \begin{cases} \alpha_1 x_t + \xi_t, & d_t = 1, \\ \alpha_2 + \eta_t, & d_t = 2. \end{cases} \quad (4)$$

- Mileage transition: Dirichlet-distributed $\eta = (\eta_1, \eta_2, \eta_3)$ (eq. 9-10, Norets 2012).
- Unobservables: DGP 1 EV-IID; DGP 2 $\eta_t \sim N(0, h_\eta^{-1})$ trunc., $\xi_t \sim N(\rho\xi_{t-1}, h_\xi^{-1})$ trunc.
- Object of approximation:

$$F(x, \xi, \theta) = \beta \sum_{j=1}^3 \eta_j (\mathbb{E}[V(x + j - 1, \xi', \eta'; \theta) \mid \xi; \theta] - EV_2(\theta)). \quad (5)$$

4.3 Reparametrization and Gibbs Sampler (Norets, 2012)

- Latent variable substitution $\eta_{t,i} \rightarrow \chi_{t,i}$ working latent variable.
- Joint posterior decomposition (eq. 5, Norets 2012).
- Gibbs blocks for $\chi_{t,i}, \xi_{t,i}, \rho, \eta, (\alpha, \beta)$; truncated normal (Geweke 1991) / Dirichlet proposals.
- $F(\cdot)$ enters every block via joint density $\rightarrow$ KASN plug-in substitutes exact DP or FFANN.

4.4 KASN Approximation of $F(\cdot)$

- Input: $\tilde{X} = (\tilde{x}, \tilde{\xi}, \tilde{\alpha}_1, \tilde{\alpha}_2, \tilde{\rho}, \tilde{\eta}_1, \tilde{\eta}_2, \tilde{\eta}_3)$ after CDF normalization $\tilde{X}_{t,j} = \hat{F}_n^j(X_{t,j})$, compact support $[0, 1]^d$.
- Sparse compositional assumption (Assmp. K.3(a)):

$$F(\tilde{X}) = \sum_{i=1}^{s_0} \Phi_i \left(\sum_{j \in \mathcal{J}_i} \phi_{i,j}(\tilde{X}_{t,j}) \right), \quad s_0 \ll 2d + 1. \quad (6)$$

Plausible: only few state-parameter inputs drive F ; latent-factor interpretation.

- Architecture parameters:

$$L_n \asymp \log n, \quad W_n \asymp s_0 \asymp \log n, \quad G_n \asymp n^\gamma, \quad (7)$$

$$\Delta_n \asymp n^{\zeta_\delta}, \quad \zeta_\delta > 0, \quad \frac{1}{4k} < \gamma < \frac{1}{2}, \quad (8)$$

$$\bar{s}_0 = s_0 \times L_n = O((\log n)^2). \quad (9)$$

- KASN sieve space:

$$\begin{aligned} \mathcal{K}_n = \left\{ f(\tilde{X}; \mathbf{c}_n) \mid f(\mathbf{x}; \mathbf{c}_n) = \Phi_{L_n} \circ (\Phi_{L_n-1} + h) \circ \cdots \circ (\Phi_1 + h) \circ \Phi_0(\mathbf{x}), \right. \\ \left. \mathbf{c}_n \in \mathbb{R}^{r_n \times G_n}, \quad \sum_{j=1}^{r_n} \|\mathbf{c}_j\|_2 \leq \Delta_n \right\}, \end{aligned} \quad (10)$$

with $h(x) = x$ residual, B-splines order $k \geq m + 1$, $\ell_{1,2}$ Group-Lasso.

- Extremum estimator (quadratic loss):

$$\hat{F}_n = \underset{f \in \mathcal{K}_n}{\arg \min} \frac{1}{n} \sum_{t=1}^n [\ell(Z_t, f) - \lambda_n \|\mathbf{c}_n\|_{1,2}], \quad (11)$$

equivalent to constrained problem $\|\hat{\mathbf{c}}(\lambda_n)\|_{1,2} = \Delta_n$.

- Training: draw inputs from prior support; solve DP at each θ^j on random grids (Rust, 1997), average to etalon; optimize via Adam with early stopping $\varepsilon_{tol} = o(n^{-1/2})$.
- Theoretical guarantee (KASN paper, Theorems 1–3):

$$\|\hat{F}_n - F\|_{L^2(\mu)} = o_P(n^{-1/4}), \quad (12)$$

uniform over $d = O(n^{\zeta_d})$. Sufficient for valid posterior inference (extends Norets 2012 Theorem 1.3).

4.5 Inference Extensions Facilitated by KASN

Extension I – Average Partial Derivatives of F

- Estimand: $\Gamma_1(\theta_0) = \mathbb{E}\left[\frac{\partial F(X_t, \theta_0)}{\partial x_j}\right]$ (structural elasticity).
- Plug-in: $\hat{\Gamma}_{1,n} = \frac{1}{n} \sum_{t=1}^n \partial \hat{F}_n(X_t, \hat{\theta}_n) / \partial x_j$.
- $\sqrt{n}$ -asymptotic normality via Riesz representer + Sobolev-norm bound (KASN paper Cor. 1, Thm. 4).

Extension II – Counterfactual Welfare Functionals

- Estimand: $\Gamma_2(\theta_0) = \mathbb{E}_\mu[V(X_t; \theta^{cf})] - \mathbb{E}_\mu[V(X_t; \theta_0)]$ (e.g. subsidy $\alpha_2^{cf} = (1 - \tau)\alpha_{2,0}$).
- Plug-in: recover $\hat{V}_n$ from $\hat{F}_n$ via (1)–(2); $\hat{\Gamma}_{2,n} = \frac{1}{n} \sum_{t=1}^n [\hat{V}_n(X_t; \theta^{cf}) - \hat{V}_n(X_t; \hat{\theta}_n)]$.
- Smooth functional $\Rightarrow \sqrt{n}$ -normal (nonlinear extension of Chen and Shen, 1998). Not available under FFANN.

Extension III – Average Marginal Effects on CCPs

- Hotz–Miller: $P(d = 2|x, \theta) = \Lambda(F(x, 2, \theta) - F(x, 1, \theta))$.
- Estimand: $\Gamma_3(\theta_0) = \mathbb{E}\left[\frac{\partial P(d=2|X_t, \theta_0)}{\partial x_j}\right]$.
- Plug-in: replace F by $\hat{F}_n$, differentiate Λ analytically. $\sqrt{n}$ -normal under Assumptions K.1–K.4.
- Dimension-free CCP elasticity, closed-form influence function.

Bayesian Reading (If time permits, probably going to leave to future work)

- Posterior functional: $\hat{\Gamma}_k^{\text{post}} = \frac{1}{M} \sum_{m=1}^M \Gamma_k(\hat{F}_n, \theta^{(m)})$.
- $o_P(n^{-1/4})$ rate ensures posterior concentration; FFANN lacks this guarantee $\rightarrow$ credible intervals not theoretically supported.

5 Conclusion

- Summary of contribution:
 - KASN as drop-in replacement for FFANN in Norets (2012); first dimension-free, $o_P(n^{-1/4})$, inference-ready value-function approximator inside MCMC for DDCMs.
 - Three new inference targets (average derivatives, counterfactual welfare, CCP marginal effects) with valid $\sqrt{n}$ -asymptotic normality and confidence intervals.
- Empirical takeaways:
 - KASN more accurate than FFANN at comparable size; MCMC posterior closer to exact DP (DGP 1).
 - Nominal coverage on all three functionals; FFANN plug-in fails.
 - Sparsity diagnostics reveal active edges (interpretable map of which inputs drive F).
- Limitations:

- Offline KASN training cost (high) vs. FFANN (trade-off against estimation-time savings).
- Sample-splitting/cross-fitting under serial dependence remains open.
- Restricted to single-agent DDCMs; dynamic games left to future work.
- Future directions:
 - Rich DDCMs (random coefficients, common shocks) – the class flagged by Norets (2012).
 - DML cross-fitting for time series (Semenova et al., 2023; Chernozhukov et al., 2018).
 - Active-edge structure as economic interpretability tool.
 - Extension to dynamic games and continuous control via conditional-expectation representation.

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